

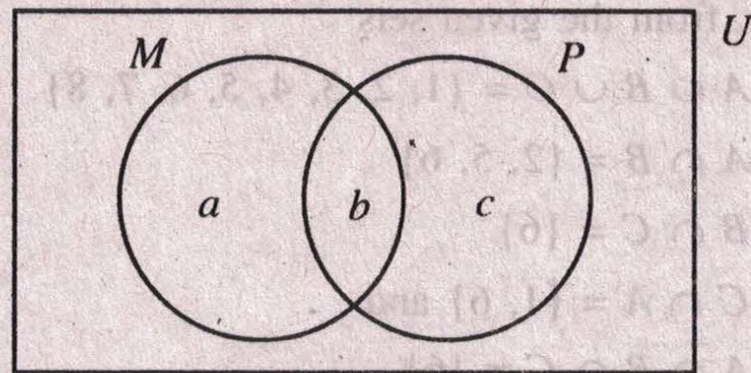
Ex. 13 : The results of students of two classes of a school with 120 students in the subject of mathematics and physics are obtained as follows :

- (i) The no. of students passing in mathematics is 55.
- (ii) The no. of students passing in physics is 60.
- (iii) The no. of students passing in both the subjects is 22.

Answer the following :

- (i) How many students are passing only in mathematics ?
- (ii) How many students are passing only in physics ?
- (iii) How many students are failing in both the subjects ?

Sol. :



Area a indicates no. of students passing only in Maths.

Area c indicates no. of students passing only in Physics.

Area b indicates no. of students passing in both the subjects.

$$\begin{aligned}\text{Now } n(M) &= \text{no. of students passing in Mathematics} \\ &= 55 = a + b \quad \dots(1)\end{aligned}$$

$$\begin{aligned}n(P) &= \text{no. of students passing in Physics} \\ &= 60 = b + c \quad \dots(2)\end{aligned}$$

$$\begin{aligned}\text{and } n(M \cap P) &= \text{no. of students passing in both} \\ &= 22 = b \quad \dots(3)\end{aligned}$$

$\therefore$ From (1), (2) and (3)

$$b = 22$$

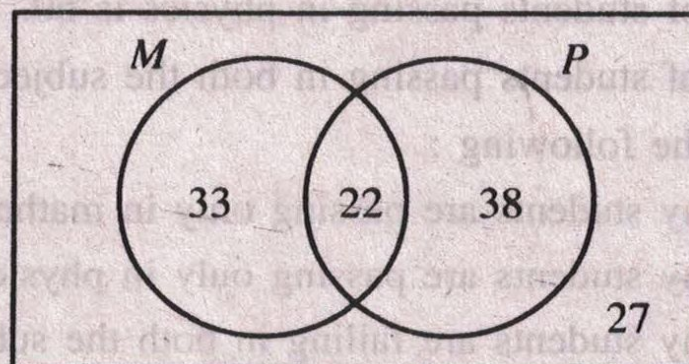
$$\therefore 22 + a = 55 \Rightarrow a = 33$$

$$\text{and } b + c = 60 \Rightarrow c = 38$$

$$\begin{aligned}\text{and } n(M \cup P) &= n(M) + n(P) - n(M \cap P) \\ &= 55 + 60 - 22 \\ &= 93\end{aligned}$$

$\therefore$ no. of students failing in both the subjects is

$$120 - 93 = 27.$$



Ex. 14 : There are 63 students in a class in a school. From these, 15 students do not play hockey or football. 35 students play football and 20 students play hockey.

Answer the following :

- (i) How many students play both the games.
- (2) How many students play only football.
- (3) How many students play only hockey.

Sol. : Let U = Total no. of students in the class

A = no. of students playing hockey

B = no. of students playing football.

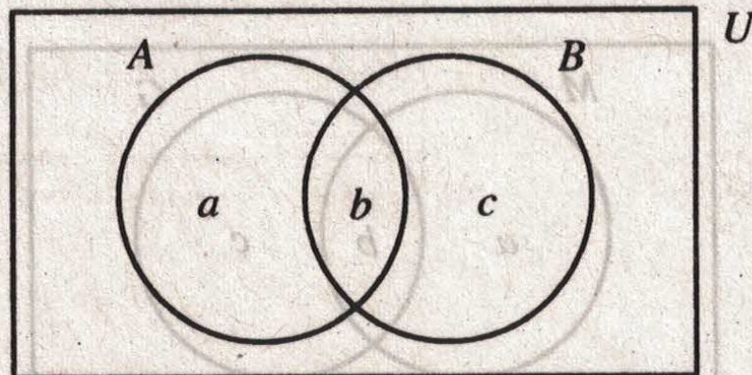
$$n(U) = 63, n(A) = 20, n(B) = 35.$$

$$n(A' \cap B') = n(A \cup B)' = 15 \quad (\because \text{Students not playing hockey and not playing football})$$

$$\therefore n(A \cup B) = n(U) - n(A \cup B)'$$

$$= 63 - 15$$

$$= 48$$



From the Venn diagram

$$\text{Students playing hockey} = a + b = 20$$

$$\text{Students playing football} = b + c = 35$$

and $a + b + c = 48$ = total no. of students playing the games A and B.

$$\therefore (a + b + c) - (a + b) = 48 - 20$$

$$\Rightarrow c = 28$$

Now, $b + c = 35$

$$\Rightarrow b = 35 - 28$$

$$b = 7$$

and $a + b = 20$

$$\Rightarrow a = 20 - 7$$

$$a = 13$$

No. of students playing only hockey is 13.

No. of students playing only football is 28.

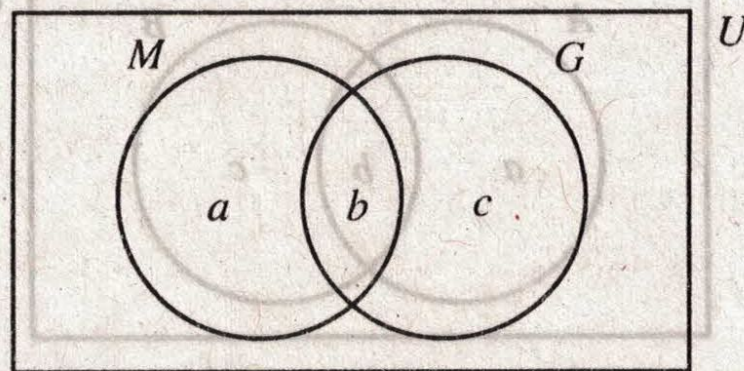
No. of students playing both hockey and football is 7.

Ex. 15 : The survey of a housing society of 250 residents in Bombay provides the following information :

- (i) Every resident can speak Gujarati or Marathi.
- (ii) 148 residents speak Gujarati.
- (iii) 175 residents speak Marathi.

Then determine how many residents of this society speak the languages and how many speak only Gujarati ?

Sol. :



$$n(M \cup G) = n(U) = 250$$

($\because$ each speaks Gujarati or Marathi)

$$\therefore a + b + c = 250$$

$$\text{People speaking Marathi} = a + b = 175$$

$$\therefore (a + b + c) - (a + b) = 250 - 175$$

$$c = 75$$

$\therefore$ People speaking only Gujarati are 75.

Now people speaking Gujarati = $b + c = 148$

$$\therefore b + c = 148$$

$$b = 148 - 75$$

$$b = 73$$

$\therefore$ People speaking both the languages is 73.

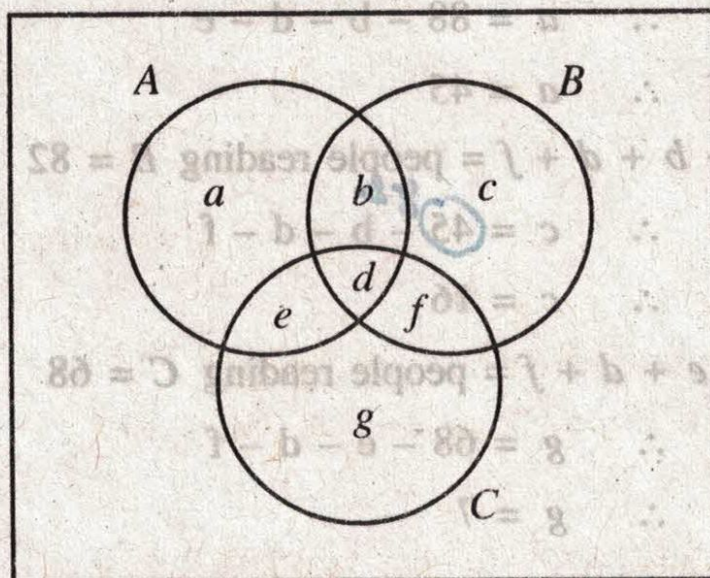
Ex. 16 : Three newspaper A, B, C are published from a city. The survey of 200 people in the city provided the following information :

- (1) 88 people read A.
- (2) 82 people read B.
- (3) 68 people read C.
- (4) 28 people read both A and B.
- (5) 46 people read both B and C.
- (6) 23 people read both C and A.
- (7) 8 people read all the three papers.

Answer the following :

- (1) Out of 200 people, how many are reading at least one paper ?
- (2) How many people read none of these papers ?
- (3) What is the number of people who read exactly two papers ?

Sol. :



From the figure

a = people who read only paper A

b = people who read both A and B but not C.

c = people who read only paper B.

d = people who read all the three papers.

e = people who read both A and C but not B.

f = people who read both B and C but not A.

g = people who read only paper C.

$$\therefore d = 8 \quad \dots(1)$$

$$b + d = \text{people reading both A and B} = 28$$

$$\therefore b = 28 - 8 = 20$$

$$\therefore b = 20 \quad \dots(2)$$

$$f + d = \text{people reading both B and C} = 46$$

$$\therefore f = 46 - 8$$

$$\therefore f = 38 \quad \dots(3)$$

$$e + d = \text{people reading both A and C} = 23$$

$$\therefore e = 23 - 8$$

$$\therefore e = 15 \quad \dots(4)$$

$$\therefore \text{Now } a + b + d + e = \text{people reading A} = 88$$

$$\therefore a = 88 - b - d - e$$

$$\therefore a = 45 \quad \dots(5)$$

$$\text{Also } c + b + d + f = \text{people reading B} = 82$$

$$\therefore c = 82 - b - d - f$$

$$\therefore c = 16 \quad \dots(6)$$

$$\text{and } g + e + d + f = \text{people reading C} = 68$$

$$\therefore g = 68 - e - d - f$$

$$\therefore g = 7 \quad \dots(7)$$

Now,

no. of people reading atleast one paper is

$$a + b + c + d + e + f + g = 149$$

no. of people not reading only paper

$$= 200 - (\text{no. of people reading atleast one paper})$$

$$= 200 - 149$$

$$= 151$$

no. of people reading exactly one paper

$$= a + c + g$$

$$= 45 + 16 + 7$$

$$= 68$$

no. of people reading exactly two papers

$$= b + e + f$$

$$= 20 + 15 + 38$$

$$= 73.$$

Ex. 17 : If $A = \{x \in \mathbb{Z} / x^2 < 49\}$

$B = \{x \in \mathbb{N} / (x + 1)^2 < 57\}$ and

$C = \{x \in \mathbb{Z} / -3 < x < 11\}$ verify

$$(i) \quad (A \cap C) \Delta (B \cap A) = (B \Delta C) \cap A$$

$$(ii) \quad (A \cup B) - C = (A - C) \cup (B - C).$$

Sol. : $A = \{-6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$

$B = \{1, 2, 3, 4, 5, 6\}$

$C = \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

(i) Firstly, we find

$$A \cap C = \{-2, -1, 0, 1, 2, 3, 4, 5, 6\}$$

$$B \cap A = \{1, 2, 3, 4, 5, 6\}$$

$$B \Delta C = (B \cup C) - (B \cap C)$$

$$= \{-2, -1, 0, 1, \dots, 10\} = \{1, 2, 3, 4, 5, 6\}$$

$$= \{-2, -1, 0, 7, 8, 9, 10\}$$

Now

$$\begin{aligned}(A \cap C) \Delta (B \cap A) &= [(A \cap C) \cup (B \cap A)] \\ &\quad - [(A \cap C) \cap (B \cap A)] \\ &= \{-2, -1, 0, 1, 2, 3, 4, 5, 6\} \\ &\quad - \{1, 2, 3, 4, 5, 6\} \\ &= \{-2, -1, 0\} \quad \dots(1)\end{aligned}$$

$$\text{and } (B \Delta C) \cap A = \{-2, -1, 0\} \quad \dots(2)$$

$$\therefore (A \cap C) \Delta (B \cap A) = (B \Delta C) \cap A$$

$$(ii) \quad A \cup B = \{-6, -5, -4, \dots, 6\}$$

$$A - C = \{-6, -5, -4, -3\}$$

$$B - C = \phi$$

$$\therefore (A - C) \cup (B - C) = \{-6, -5, -4, -3\} \quad \dots(1)$$

$$(A \cup B) - C = \{-6, -5, -4, -3\} \quad \dots(2)$$

$\therefore$ From (1) and (2)

$$(A \cup B) - C = (A - C) \cup (B - C).$$

Ex. 18 : If $U = \{-3, -1, 0, 1, 3\}$, $A = \{-3, -1, 1\}$, $B = \{-1, 1, 3\}$ and $C = \{-1, 0, 1\}$ then verify the following

$$(1) \quad (A \cup B)' = A' \cap B'$$

$$(2) \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(3) \quad B - A = A' \cap B = B - (A \cap B)$$

$$(4) \quad A \Delta B = (A - B) \cup (B - A)$$

$$(5) \quad (B')' = B$$

$$(6) \quad A \cup B = A \cup [B - (A \cap B)].$$

Sol. :

$$(1) \quad (A \cup B)' = A' \cap B'$$

$$\text{Now, } U = \{-3, -1, 0, 1, 3\}$$

$$A = \{-3, -1, 1\}$$

$$B = \{-1, 1, 3\}$$

$$A \cup B = \{-3, -1, 1, 3\}$$

$$\therefore (A \cup B)' = \{0\} \quad \dots(1)$$

$$\text{And } A' = \{0, 3\}, B' = \{-3, 0\}$$

$$\therefore A' \cap B' = \{0\} \quad \dots(2)$$

$\therefore$ From (1) and (2)

$$(A \cup B)' = A' \cap B'$$

$$(2) \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$\text{Here } B \cap C = \{-1, 1\}$$

$$\therefore A \cup (B \cap C) = \{-3, -1, 1\} \quad \dots(1)$$

$$A \cup B = \{-3, -1, 1, 3\}$$

$$A \cup C = \{-3, -1, 0, 1\}$$

$$\therefore (A \cup B) \cap (A \cup C) = \{-3, -1, 1\} \quad \dots(2)$$

$\therefore$ From (1) and (2),

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(3) \quad B - A = A' \cap B = B - (A \cap B)$$

$$\text{Here } B - A = \{3\}$$

$$A' = \{0, 3\}$$

$$\therefore A' \cap B = \{3\}$$

$$\text{and } A \cap B = \{-1, 1\}$$

$$\therefore B - (A \cap B) = \{3\}$$

$\therefore$ From (1), (2), (3)

$$B - A = A' \cap B = B - (A \cap B)$$

$$(4) \quad A \Delta B = (A - B) \cup (B - A)$$

$$\text{Here } A \cup B = \{-3, -1, 1, 3\}$$

$$A \cap B = \{-1, 1\}$$

$$\therefore A \Delta B = (A \cup B) - (A \cap B) = \{-3, 3\}$$

$$\text{and } A - B = \{-3\}$$

$$B - A = \{3\}$$

$$\therefore (A - B) \cup (B - A) = \{-3, 3\}$$

$$(5) \quad (B')' = B$$

$$B = \{-1, 1, 3\} \Rightarrow B' = \{-3, 0\}$$

$$\therefore (B')' = \{-1, 1, 3\}$$

$$\therefore (B')' = B.$$

$$(6) \quad A \cup B = A \cup [B - (A \cap B)]$$

$$\text{Here } A \cup B = \{-3, -1, 1, 3\}$$

....(1)

$$\text{and } A \cap B = \{-1, 1\}$$

$$\therefore B - (A \cap B) = \{3\}$$

$$\therefore A \cup [B - (A \cap B)] = \{-3, -1, 1, 3\}$$

....(2)

$$\therefore \text{From (1) and (2)}$$

$$A \cup B = A \cup [B - (A \cap B)].$$

Ex. 19 : $A = \{5, 6, 7\}$, $B = \{7, 8\}$, $C = \{5, 8\}$ then verify

$$(i) \quad A \times (B \cap C) = (A \times B) \cap (A \times C)$$

$$(ii) \quad A \times (B - C) = (A \times B) - (A \times C)$$

$$(iii) \quad A \times (B \Delta C) = (A \times B) \Delta (A \times C).$$

Sol. :

$$(i) \quad A \times (B \cap C) = (A \times B) \cap (A \times C)$$

$$\text{Here } B \cap C = \{8\}$$

$$\therefore A \times (B \cap C) = \{(5, 8), (6, 8), (7, 8)\} \quad \text{....(1)}$$

$$A \times B = \{(5, 7), (5, 8), (6, 7), (6, 8), (7, 7), (7, 8)\}$$

$$A \times C = \{(5, 5), (5, 8), (6, 5), (6, 8), (7, 5), (7, 8)\}$$

$$\therefore (A \times B) \cap (A \times C) = \{(5, 8), (6, 8), (7, 8)\} \quad \text{....(2)}$$

$$\therefore \text{From (1) and (2) the result follows.}$$

$$(ii) \quad B - C = \{7\}$$

$$\therefore A \times (B - C) = \{(5, 7), (6, 7), (7, 7)\} \quad \text{....(1)}$$

$$A \times B = \{(5, 7), (5, 8), (6, 7), (6, 8), (7, 7), (7, 8)\}$$

$$A \times C = \{(5, 5), (5, 8), (6, 5), (6, 8), (7, 5), (7, 8)\}$$

$$\therefore (A \times B) - (A \times C) = \{(5, 7), (6, 7), (7, 7)\} \quad \text{....(2)}$$

$$\therefore \text{From (1) and (2) the result follows.}$$

(iii) $A \times (B \Delta C) = (A \times B) \Delta (A \times C)$

Here $B \cup C = \{5, 7, 8\}$

$B \cap C = \{8\}$

Now $B \Delta C = (B \cup C) - (B \cap C) = \{5, 7\}$

$\therefore A \times (B \Delta C) = \{(5, 5), (5, 7), (6, 5), (6, 7),$
 $(7, 5), (7, 7)\} \dots(1)$

$A \times B = \{(5, 7), (5, 8), (6, 7), (6, 8), (7, 7), (7, 8)\}$

$A \times C = \{(5, 5), (5, 8), (6, 5), (6, 8), (7, 5), (7, 8)\}$

$\therefore (A \times B) \cup (A \times C) = \{(5, 5), (5, 7), (5, 8), (6, 5),$
 $(6, 7), (6, 8), (7, 5), (7, 7), (7, 8)\}$

and

$(A \times B) \cap (A \times C) = \{(5, 8), (6, 8), (7, 8)\}$

Now $(A \times B) \Delta (A \times C)$

$= [(A \times B) \cup (A \times C)] - [(A \times B) \cap (A \times C)]$

$= \{(5, 5), (5, 7), (6, 5), (6, 7), (7, 5), (7, 7)\}$

$\dots(2)$

$\therefore$ From (1) and (2) the result follows.